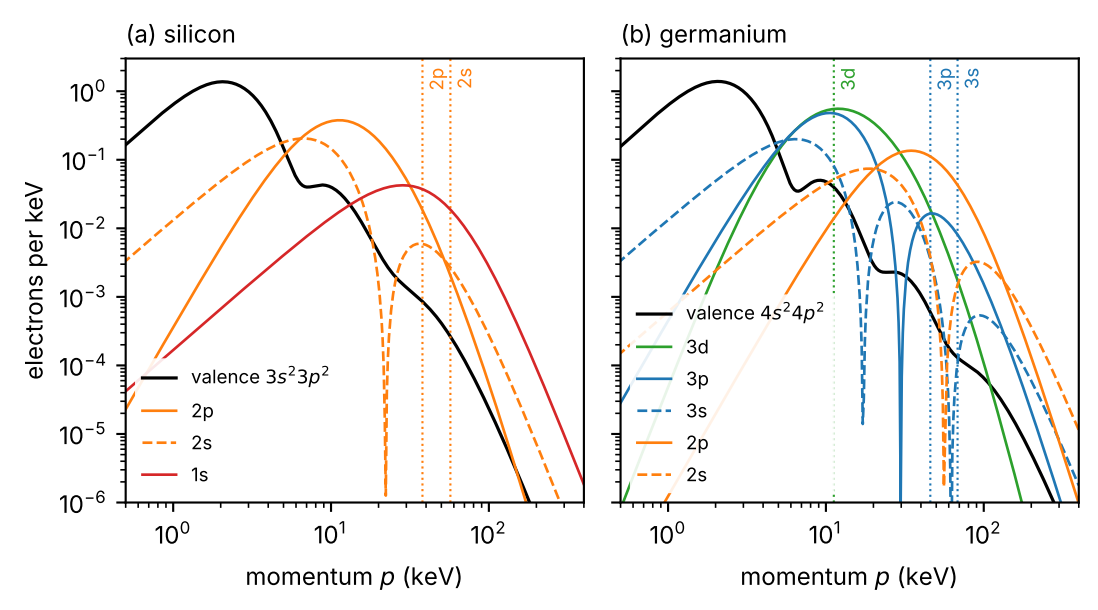
(a) silicon
(b) germanium
electrons per keV
momentum p (keV)
momentum p (keV)
valence $3s^2 3p^2$
2p
2s
1s
valence $4s^2 4p^2$
3d
3p
3s
2p
2s